Rubric

Lavoisier's ghost challenge/Measurements in the Lab

Station 1. 4 points

Level 4 (exceeds expectations) >3.5 points	**Level 3 (meets expectations) 3-3.5 points**	**Level 2 (needs improvement) 2<pts<3**	**Level 1 (does not meet expectations) 0-2 points**
Students show advanced understanding in reporting the following items (0-1 error) • Reading uncertainty • SD • RSD% • Comparison between the reported Toledo balance tolerance and their SD • Units for mass, SD and RSD • Significant figures	Students show proficient understanding in reporting the following items (2-4 errors) • Reading uncertainty • SD • RSD% • Comparison between the reported Toledo balance tolerance and their SD • Units for mass, SD and RSD • Significant figures	Students show emerging understanding in reporting the following items (up to 7 errors) • Reading uncertainty • SD • RSD% • Comparison between the reported Toledo balance tolerance and their SD • Units for mass, SD and RSD • Significant figures	Students show little or no understanding in reporting the following items (8 and more errors) • Reading uncertainty • SD • RSD% • Comparison between the reported Toledo balance tolerance and their SD • Units for mass, SD and RSD • Significant figures

Station 2. 4 points

Level 4 (exceeds expectations) **>3.5 points**	**Level 3 (meets expectations)** **3-3.5 points**	**Level 2 (needs improvement)** **2<pts<3**	**Level 1 (does not meet expectations)** **0-2 points**
Students show advanced understanding in reporting the following items (0-1 error) • Reading uncertainty/error • Relative % error • Comparison of different glassware	Students show proficient understanding in reporting the following items (2-4 errors) • Reading uncertainty/error • Relative % error • Comparison of different glassware	Students show emerging understanding in reporting the following items (up to 7 errors) • Reading uncertainty/error • Relative % error • Comparison of different glassware	Students show little or no understanding in reporting the following items (8 and more errors) • Reading uncertainty/error • Relative % error • Comparison of different glassware

Station 3. 2 points

Level 4 (exceeds expectations) **=>1.75 points**	**Level 3 (meets expectations)** **1.5-1.75 points**	**Level 2 (needs improvement)** **1<pts<1.5**	**Level 1 (does not meet expectations)** **0-1 points**
Students show advanced understanding (0-1 error) of • using the significant figures in calculations and units' conversion. • Mole concept	Students show proficient understanding (1-2 errors) of • using the significant figures in calculations and units' conversion • Mole concept	Students show emerging understanding (up to 3 errors) of • using the significant figures in calculations and units' conversion • Mole concept	Students show little or no understanding (4 and more errors) of • using the significant figures in calculations and units' conversion • Mole concept

Station 4. 4 points

Level 4 (exceeds expectations) >3.5 points	**Level 3 (meets expectations) 3-3.5 points**	**Level 2 (needs improvement) 2<pts<3**	**Level 1 (does not meet expectations) 0-2 points**
Students show advanced understanding in reporting the following items (no errors or one minor error) • Balance reading • Thermometer reading • Density graph reading • Significant figures in calculations for average mass and density • Coin material	Students show proficient understanding in reporting the following items (up to 2 errors) • Balance reading • Thermometer reading • Density graph reading • Significant figures in calculations for average mass and density • Coin material	Students show emerging understanding in reporting the following items (3 errors) • Balance reading • Thermometer reading • Density graph reading • Significant figures in calculations for average mass and density • Coin material	Students show little or no understanding in reporting the following items (4 and more errors) • Balance reading • Thermometer reading • Density graph reading • Significant figures in calculations for average mass and density • Coin material

Station 5. 2 points

Level 4 (exceeds expectations) =>1.75 points	**Level 3 (meets expectations) 1.5-1.75 points**	**Level 2 (needs improvement) 1<pts<1.5**	**Level 1 (does not meet expectations) 0-1 points**

Students show advanced understanding in reporting the following items (0-1 error) • the analog and digital scale readings • converting the temperature scales and reporting them with the correct number of significant figures	Students show advanced understanding in reporting the following items (1-2 errors) • the analog and digital scale readings • converting the temperature scales and reporting them with the correct number of significant figures	Students show emerging understanding in reporting the following items (up to 3 errors) • the analog and digital scale readings • converting the temperature scales and reporting them with the correct number of significant figures	Students show little or no understanding in reporting the following items (4 and more errors) • the analog and digital scale readings • converting the temperature scales and reporting them with the correct number of significant figures

Station 6. 4 points

Level 4 (exceeds expectations) **>3.5 points**	**Level 3 (meets expectations)** **3-3.5 points**	**Level 2 (needs improvement)** **2<pts<3**	**Level 1 (does not meet expectations)** **0-2 points**
Students show advanced understanding in reporting the following items (no errors or one minor error) • Identify which law is stated on the puzzle • An example of an experiment Lavoisier used to arrive to this law.	Students show proficient understanding in reporting the following items (up to 2 errors) • Identify which law is stated on the puzzle • An example of an experiment Lavoisier used to arrive to this law.	Students show emerging understanding in reporting the following items (3 errors) • Identify which law is stated on the puzzle • An example of an experiment Lavoisier used to arrive to this law.	Students show little or no understanding in reporting the following items (4 and more errors) • Identify which law is stated on the puzzle • An example of an experiment Lavoisier used to arrive to this law.